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Design Lab-I Project Report

FORCE SENSOR USING MAGNETOMETER

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Introduction

This work presents the design and development of a soft force sensing system based on magnetic field variation induced by deformation of an elastic structure. Conventional force sensors such as load cells and strain gauges are often rigid and unsuitable for applications requiring compliance, such as soft robotics and tactile sensing. To address this limitation, a deformable sensing mechanism using thermoplastic polyurethane (TPU) is proposed.

The sensing principle relies on the coupling between mechanical deformation and magnetic field variation. When an external force is applied to the TPU structure, it undergoes deformation, resulting in displacement of an embedded triangular permanent magnet relative to a fixed magnetometer. This displacement alters the magnetic field intensity measured by the sensor. The magnetic field is captured using the MLX90393 three-axis magnetometer, interfaced with an ESP32 microcontroller for real-time data acquisition.

Experimental characterization was carried out using a Universal Testing Machine (UTM) to apply controlled forces. Corresponding deformation and magnetic field data were recorded simultaneously. Due to the nonlinear elastic behaviour of TPU and the nonlinear dependence of magnetic field on distance, the relationship between force and magnetic field was observed to be nonlinear. A calibration model was developed by mapping the measured magnetic field values to the applied force using curve fitting techniques.

The results demonstrate that the proposed system can effectively estimate applied force based on magnetic field measurements, with reasonable sensitivity and repeatability within the tested range. The study highlights the feasibility of using soft materials combined with magnetic sensing for developing low-cost, flexible force sensors.

Theory

Magnetic Field Fundamentals:

A permanent magnet, regardless of its shape, can be modelled at sufficient distances as a **magnetic dipole** — the simplest non-trivial source of magnetic field. The dipole is characterised by its **magnetic dipole moment vector**:

$\mathbf{m} = M \cdot V \cdot \hat{\mathbf{n}}$

where:

- M is the remanent magnetisation of the magnet material (A/m)
- V is the volume of the magnet
- $\hat{\mathbf{n}}$ is the unit vector along the magnetisation axis (the direction from south to north pole within the magnet)

Sensor working (MLX90393):

It senses magnetic fields through three Hall plates, one per axis, which detect flux density in millitesla (mT). The signals undergo amplification (programmable gain), analog-to-digital conversion (17.4-bit ADC, trimmed to 16-bit output), and optional filtering/digital processing before transmission.

The Triaxis setup provides sensitivity from 0.16 µT/LSB to 6.2 µT/LSB (XY) or 0.29 µT/LSB to 11.7 µT/LSB (Z), with a wide range up to ±50 mT before saturation, ideal for strong magnets unlike Earth's field (~0.05 mT).

It has a range of 100µT up to ±50 mT.

Hall Effect:

Charged carriers (electrons or holes) flowing through the material experience a Lorentz force from the magnetic field, deflecting them to one side and creating charge separation. This builds an electric field that opposes further deflection, resulting in a measurable transverse voltage *Vh=IB/qnd*,

where I is current, *B* is magnetic flux density, *q* is carrier charge, *n* is carrier density, and *d* is thickness.

In n-type semiconductors, electrons accumulate on one side (negative Hall voltage); p-type yields the opposite.

This generate the Hall voltage across the top-bottom faces.

Key derivations :

$$F = kx, \qquad k = \frac{F}{x}$$

The equation F = kx is Hooke law for a linear elastic element. TPU is not perfectly linear like a metal spring, but for small strain and short loading time it can be approximated as a spring. In a small deformation range, the slope of the force-displacement curve is almost constant. This slope is called stiffness k.

$$\varepsilon = \frac{x}{L}, \quad \sigma = E\varepsilon, \quad F = \sigma A = \frac{EA}{L}x$$

For a simple axial member, stiffness becomes k = EA/L, where E is the tangent Young modulus of TPU in the selected strain range, A is effective loaded area, and L is effective deformation length. Your actual TPU geometry may bend, compress, shear, or combine all three. Therefore, k should be experimentally found from the slope of a known-force versus displacement test.

Important: F = kx is the first operating model. For high accuracy, use an effective stiffness k_eff from calibration.

TPU shows hysteresis, creep, and stress relaxation, so loading speed and waiting time should be kept consistent.

Equilateral triangular magnet geometry

$$A_\Delta = \frac{\sqrt{3}}{4}a^2, \qquad R_{eq} = a\sqrt{\frac{\sqrt{3}}{4\pi}}$$

For an equilateral triangular magnet of side length a, the pole face area is A_triangle = sqrt(3)a^2/4. A near-field exact analytical expression over a triangle is mathematically complex. For a usable closed-form design equation, the triangular pole is replaced by a circular pole with the same area. This keeps total pole area

unchanged and gives a better near-field expression than the point dipole model.

Equivalent-area radius R_eq is used only for analytical modeling. Final calibration still corrects the remaining shape error caused by triangular corners and edge effects.

Why the dipole model is not used

The point-dipole equation B proportional to 1/r^3 is valid only when the sensor distance is much larger than the magnet size. In our design, the magnetometer is close to the magnet. This is the near-field region, where the finite pole area, magnet thickness, and edge effects are important. Therefore, using a dipole equation as the

main derivation would give wrong physical justification.

<u>Magnetic surface charge model</u>

A uniformly magnetized permanent magnet can be represented by magnetic surface charge density sigma_m on its pole faces. The north face carries +sigma_m and the south face carries -sigma_m. Outside the magnet, the field can be calculated similarly to the electric field of charged sheets, with magnetic constants. The

magnetometer reads one component of this vector field, usually along its sensitive axis.

$$B_z(g) = \frac{\mu_0 \sigma_m}{4\pi} \iint_A \left[\frac{g}{(g^2 + \rho^2)^{3/2}} - \frac{g+t}{((g+t)^2 + \rho^2)^{3/2}} \right] dA$$

Here g is the distance from magnetometer to the near pole face,

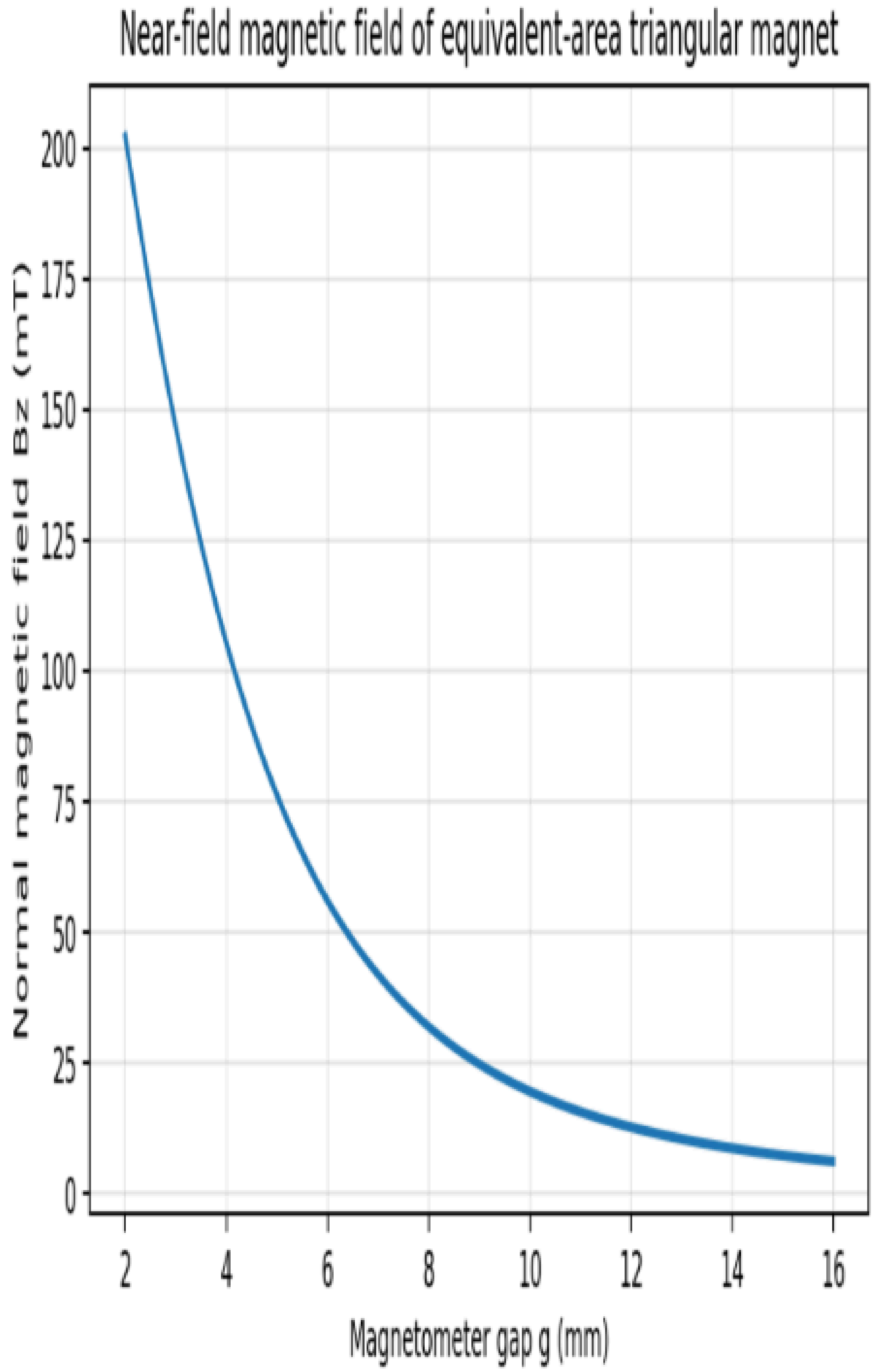
t is magnet thickness,

rho is the radial distance of a small area element from the axis, and

A is the triangular pole face.

The first term is the contribution of the near pole; the second term is the opposite contribution of the far pole. This equation is the rigorous starting point for near-field modeling. For the true triangular magnet, the integration area A is the equilateral triangle. If the centroid is used as origin, the limits are piecewise linear and depend on the chosen coordinate orientation. Because of the triangular boundary, a simple compact closed form is not convenient. Numerical integration or finite element analysis gives the most accurate result.

Magnetic Field and distance Relationship:

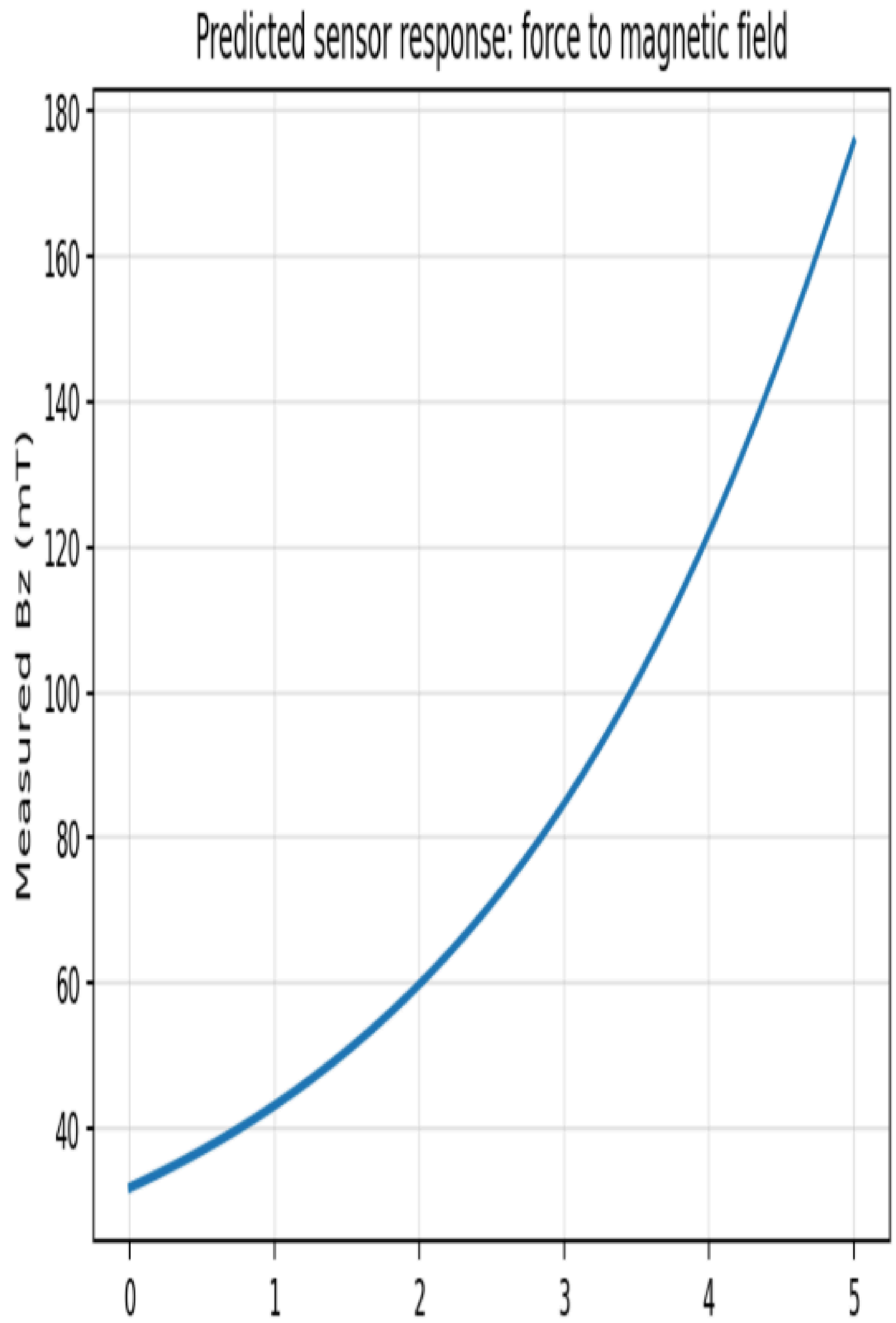


Force and magnetic field relation:
Let g0 be the initial gap at zero applied force. If the magnet moves toward the magnetometer by displacement x, then g = g0 - x. For the small-strain TPU region, x = F/k. Substituting this into the magnetic field equation

gives the theoretical sensor response.

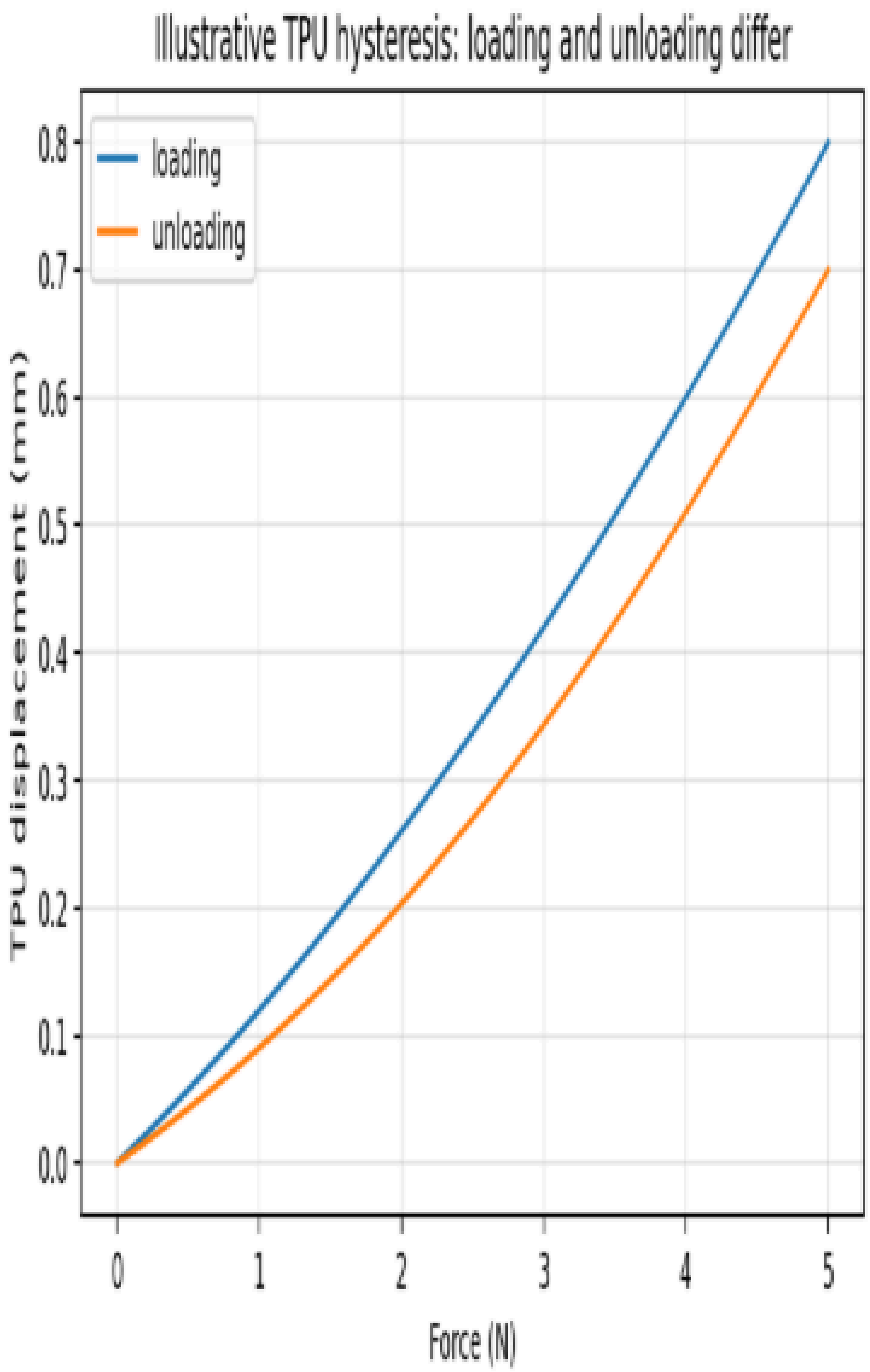
$$B_Z(F) = \frac{B_r}{2}\left[\frac{g_0 - F/k + t}{\sqrt{(g_0 - F/k + t)^2 + R_{eq}^2}} - \frac{g_0 - F/k}{\sqrt{(g_0 - F/k)^2 + R_{eq}^2}}\right]$$

This equation shows that the sensor response is nonlinear. The nonlinearity is not an error; it is the natural result of near-field magnetics. The response becomes more sensitive when the magnet gets closer because the magnetic field changes faster with gap.



TPU nonlinearity and hysteresis

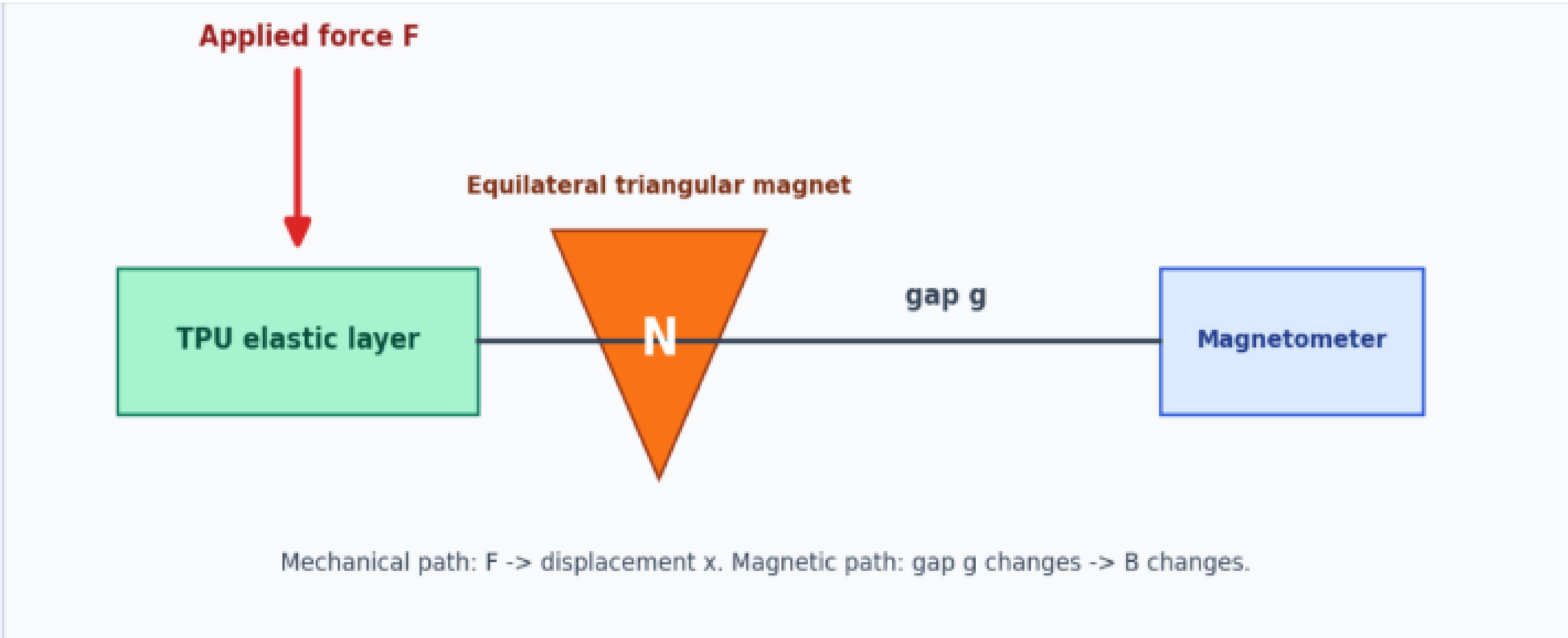
TPU is viscoelastic. It stores energy like an elastic material but also dissipates energy like a viscous material.
Therefore, the displacement under a given force may depend on loading history. Loading and unloading paths
are not exactly the same. This is called hysteresis.



Proposed Sensing Concept

The proposed sensing system is based on converting applied force into measurable magnetic field variation through a deformable structure. The core idea is to use a **TPU-based**

elastic Mold with an embedded **triangular permanent magnet**, positioned relative to a fixed MLX90393 magnetometer.



When an external force is applied to the TPU structure, it undergoes deformation. This deformation causes a change in the position of the embedded magnet with respect to the sensor. Since the magnetic field measured by the sensor depends on the relative position and orientation of the magnet, any displacement results in a variation in the magnetic field values (Bx ,By ,Bz). These changes are captured by the magnetometer and processed using the ESP32.

In this way, the applied force is not measured directly but is inferred indirectly through magnetic field variation. By collecting corresponding force and magnetic field data, a calibration relationship is established to estimate force from sensor readings.

This approach works because it effectively links three physical phenomena:

- Mechanical deformation of TPU under applied force
- Displacement of the magnet due to deformation

- Variation of magnetic field with respect to distance and orientation

Each stage in this chain is sensitive to changes, allowing small forces to produce measurable variations in magnetic field. The magnetometer provides high-resolution measurements, enabling accurate detection of these changes. Additionally, since the sensing is contactless at the electrical level (no direct strain measurement), it reduces issues like material fatigue affecting electrical properties

What makes this approach different

The proposed system differs from conventional force sensing methods in several ways:

- It uses a **soft and deformable TPU structure**, making it suitable for applications requiring flexibility.
- The sensing is based on **magnetic field variation rather than electrical resistance or capacitance**, reducing dependency on material electrical properties
- The use of a **triangular magnet** introduces non-uniform magnetic field distribution, which can improve sensitivity to displacement
- The system is relatively **simple and low-cost**, using commonly available components like ESP32 and MLX90393

Overall, this approach provides a flexible and adaptable method for force sensing, especially in scenarios where traditional rigid sensors are not suitable.

PROCEDURE

1. Design And Development of Sensor

The design and development of the sensor involved creating a deformable structure, selecting appropriate component placement, and ensuring that the system produces measurable and consistent magnetic field variation under applied force.

Structural Design of Tpu Mold

The sensor structure was developed using a 3D-printed TPU Mold, chosen for its flexibility and ability to undergo significant deformation under applied force. The geometry of the Mold was designed to allow controlled compression when force is applied.

Initially, simple shapes were considered, but it was observed that deformation was not uniform across the structure. Therefore, the design was adjusted to ensure that deformation is more predictable and concentrated in the region where the magnet is placed. This helps in achieving a consistent relationship between applied force and displacement.

The thickness and internal structure of the Mold were also important factors. A balance was required:

- Too stiff → very small deformation → low sensitivity
- Too soft → excessive deformation → unstable readings

Hence, the design was finalized to provide measurable deformation within the operating force range.

Magnet Geometry and Placement

A triangular permanent magnet was used in the design. Unlike symmetric shapes (such as circular magnets), a triangular magnet produces a more non-uniform magnetic field, which improves sensitivity to positional changes.

The magnet was embedded inside the TPU Mold at a specific location such that its movement under deformation directly affects the magnetic field at the sensor. Care was taken to ensure:

- The magnet is securely fixed inside the structure
- Its orientation remains consistent during deformation

The relative distance between the magnet and sensor was chosen based on trial observations to ensure that the magnetic field variation falls within the measurable range of the sensor.

Sensor positioning and alignment

The MLX90393 magnetometer was placed at a fixed position relative to the TPU structure. Proper alignment between the sensor and magnet was critical for accurate measurements.

During development, it was observed that:

- Small misalignments significantly affect readings
- Off-axis positioning reduces sensitivity

Therefore, the setup was adjusted to:

- Keep the sensor aligned with the primary direction of magnet movement
- Maintain a stable and fixed mounting to avoid vibrations or shifts

The final configuration ensured that deformation of the TPU structure results in a clear and consistent change in the magnetic field measured by the sensor.

The overall design focused on:

- Achieving controlled and repeatable deformation
- Avoid structural failure like Bucking when dealing with large forces
- Maximizing sensitivity of magnetic field variation
- Ensuring stable and aligned sensor–magnet configuration

Through iterative adjustments in geometry, magnet placement, and alignment, a functional sensor structure was successfully developed.

2. Implementation and hardware setup

Esp32 and mlx90393 interfacing

The MLX90393 magnetometer was interfaced with the ESP32 microcontroller using the I²C communication protocol. The sensor provides three-axis magnetic field data (Bx ,By ,Bz), which are read by the ESP32.

The basic connections included:

- Power supply (3.3 VCC and GND) to the sensor
- I²C communication lines:
    - SDA (data line)
    - SCL (clock line)

The ESP32 was programmed to initialize the sensor, configure its measurement parameters, and continuously read magnetic field data. Appropriate delay and sampling settings were used to ensure stable readings.

During implementation, attention was given to:

- Proper soldering of connections to avoid signal loss
- Stable power supply to prevent fluctuations
- Correct I²C addressing and communication setup

Successful interfacing was confirmed by observing consistent magnetic field readings when the magnet was moved manually.

Assembly of complete system

After successful interfacing, the complete system was assembled by integrating the mechanical and electronic components.

The TPU Mold with the embedded triangular magnet was positioned relative to the MLX90393 sensor such that any deformation in the structure results in measurable magnetic field variation. The sensor was fixed in a stable position on a acrylic sheet of 4mm thickness to avoid any unintended movement during testing.

Key considerations during assembly included:

- Maintaining fixed relative positioning between magnet and sensor
- Ensuring the magnet remains properly embedded and does not shift
- Providing mechanical stability to the entire setup

The ESP32 and sensor were mounted securely to minimize vibrations and external disturbances. The overall assembly was arranged such that force could be applied vertically without disturbing the alignment.

Data acquisition setup

The data acquisition system was designed to record magnetic field values corresponding to applied force.

The ESP32 continuously reads the magnetic field components (B_x ,B_y ,B_z) from the sensor and transmits the data to a computer for storage and analysis. Data was typically monitored using a serial monitor interface.

During experiments:

- Force was applied using a Universal Testing Machine (UTM)
- At each force level, corresponding magnetic field readings were recorded
- Multiple readings were taken to ensure consistency

To improve reliability:

- Readings were taken after allowing the system to stabilize
- Noise was minimized by keeping the setup stable
- Data was recorded in a structured format for further processing

This setup enabled synchronized collection of force and magnetic field data, which was later used for calibration and analysis.

Design challenges and iterative improvements

- Problems in geometry
- Issues in sensitivity
- Changes made in design
- Why those changes improved performance

In the initial design, the geometry of the TPU Mold did not produce uniform deformation under applied force. When force was applied at different points, the deformation varied significantly, leading to inconsistent displacement of the magnet. As a result, the magnetic field readings were not repeatable.

Another issue observed was that deformation was not always directed along a single axis. Instead, the structure deformed in multiple directions, causing unpredictable movement of the magnet relative to the sensor.

Improvement made:
The geometry of the Mold was modified to make deformation more controlled and concentrated in a specific region. The design was adjusted to guide deformation primarily along the axis aligned with the sensor. This helped in ensuring that the magnet displacement remained consistent for a given applied force.

Changes made in design

Based on the observed issues, several modifications were implemented:

- Adjustment of TPU Mold geometry to improve deformation behaviour
- Optimization of magnet placement to ensure maximum displacement effect
- Refinement of sensor alignment to maintain consistent readings
- Strengthening of mounting to avoid unwanted movement or vibrations

Each modification was tested experimentally, and the design was progressively refined through multiple iterations.

Improvements in performance

The design changes led to significant improvements in the performance of the sensor:

- **Better repeatability:** Similar force values produced consistent magnetic field readings
- **Increased sensitivity:** Small changes in force resulted in noticeable changes in sensor output
- **Improved stability:** Reduced variation due to better alignment and mounting
- **More predictable behavior:** Controlled deformation resulted in smoother and more reliable data trends

Overall, the iterative design process helped in transforming an initially unstable system into a functional and reliable force sensing setup.

Experimental methodology

Force application using UTM

A Universal Testing Machine (UTM) was used to apply controlled and measurable force on the TPU-based sensor structure. The sensor assembly was placed on the base platform of the UTM, ensuring proper alignment between the loading head and the deformable region of the mold.

Force was applied gradually in a controlled continuous manner with a deformation rate of 0.5mm/minute to avoid sudden impacts and to allow stable deformation of the TPU material. The loading was performed in a continuous manner within a defined force range suitable for the designed structure.

Care was taken to:

- Ensure that force was applied vertically and centrally
- Avoid lateral movement or tilting of the setup
- Maintain consistent contact between the UTM head and the sensor surface

This setup allowed accurate measurement of the applied force corresponding to each level of deformation.

Measurement procedure

The experiment was conducted by simultaneously observing the applied force and the corresponding magnetic field values.

The procedure followed was:

1. The sensor system was initialized and allowed to stabilize
2. Initial magnetic field readings were recorded without any applied force
3. Force was applied incrementally using the UTM
4. Corresponding magnetic field values (B_x ,B_y ,B_z) were recorded
5. The process was repeated for multiple force levels within the operating range

To ensure reliability:

- Measurements were repeated for multiple trials
- Sudden loading/unloading was avoided
- Consistent conditions were maintained throughout the experiment

Data recording strategy

The MLX90393 sensor provided three-axis magnetic field data, which was read by the ESP32 and transmitted to a computer via serial communication.

The data recording strategy included:

- Logging of Bx ,By ,Bz values at each force step
- Recording corresponding force values from the UTM
- Organizing the data in tabular format for analysis

In some cases, the magnetic field magnitude was also calculated using:

B^=Bx^2 +By^2 +Bz^2

This helped in reducing directional dependency and simplifying analysis.

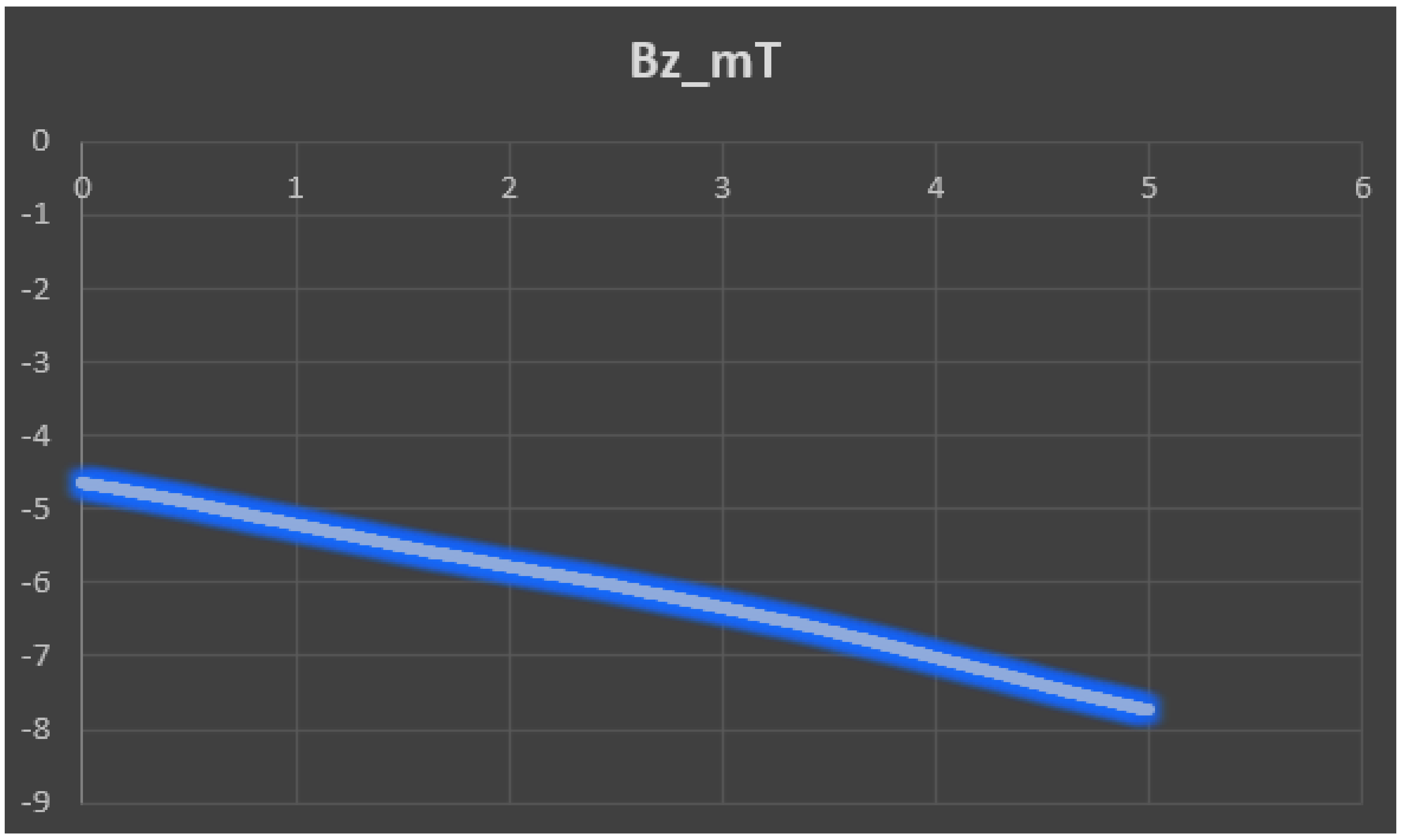
To improve data quality:

- Multiple readings were taken at each force level and averaged
- Data was recorded only after stabilization

The collected dataset formed the basis for further analysis and calibration of the sensor.

OBSERVATIONS

Force vs deformation curve



The plotted force–deformation curve represents the mechanical response of the TPU structure under applied load. From the graph, it is observed that the relationship between force and deformation is **nonlinear**, especially at higher deformation values.

<u>Initial linear region :</u>

At small deformation (approximately from 0 to ~2 units), the curve is nearly linear. In this region, the material behaves approximately like a linear elastic material, where force is proportional to deformation.

This indicates that the TPU initially resists deformation uniformly, and stiffness remains approximately constant.

<u>Non-linear region</u>

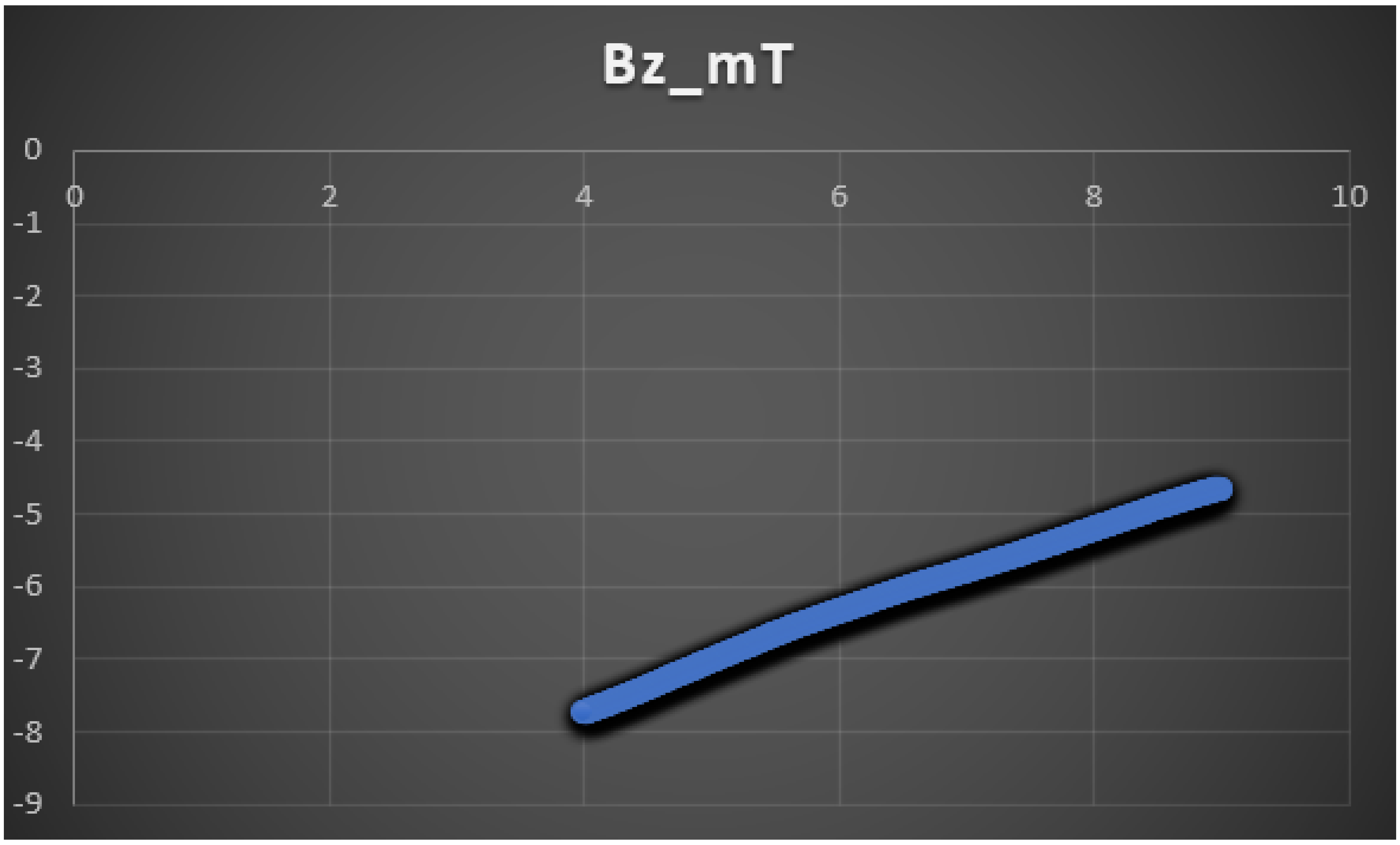
As deformation increases (beyond ~2 units), the curve starts deviating from linearity and becomes steeper. This indicates that:

- The stiffness of the TPU increases with deformation
- More force is required for the same increment in deformation

This behaviour is typical of elastomeric materials like TPU and occurs due to:

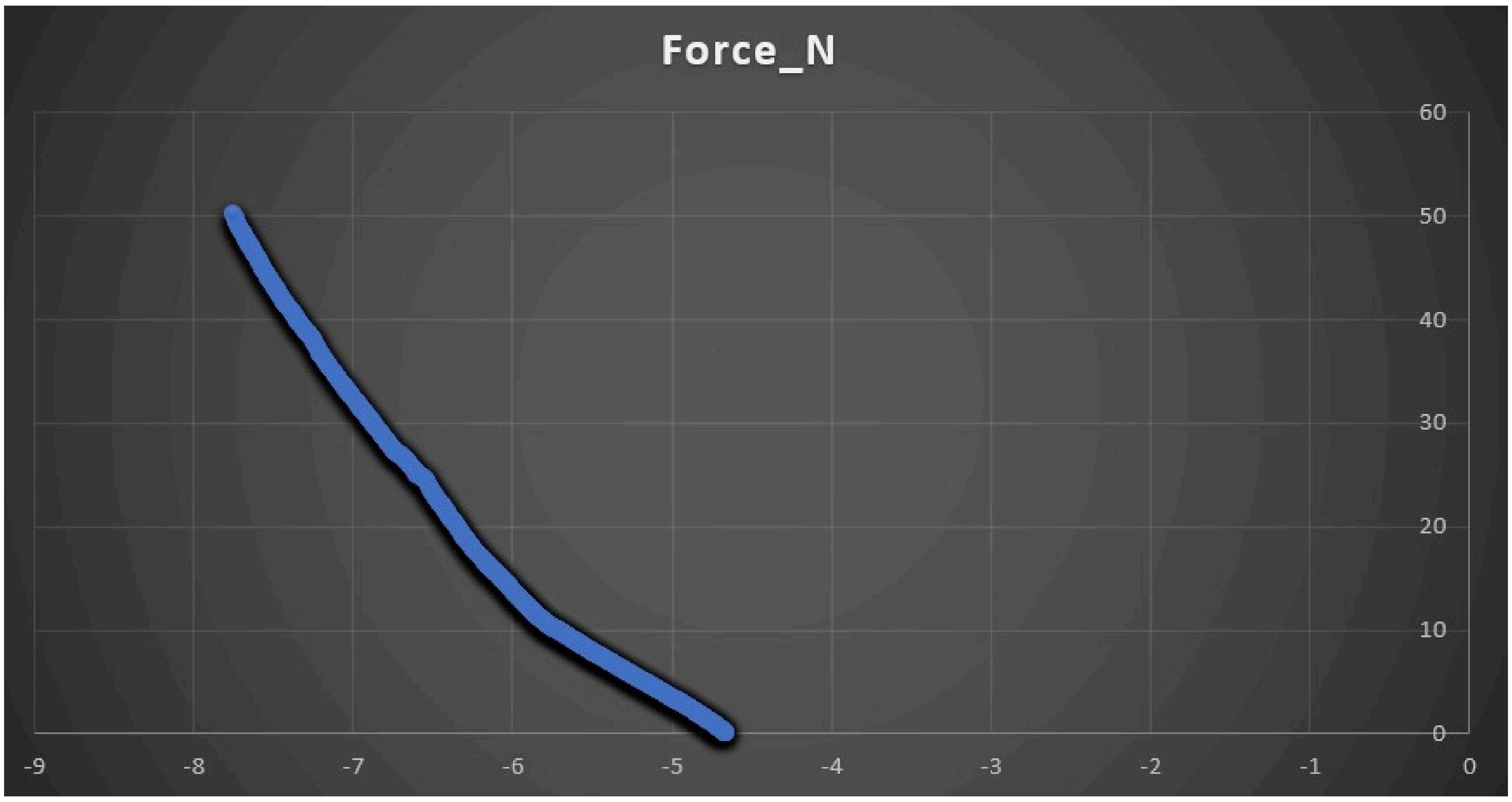
- Internal molecular chain alignment
- Reduction in available free deformation space
- Geometric effects of the structure

Magnetic Field vs Distance



RESULTS AND ANALYSIS AND VERIFICATION OF DATA

Force vs Magnetic Field Mapping



Error analysis and limitations

The performance of the developed force sensor is influenced by multiple factors arising from material behaviour, sensor characteristics, and experimental conditions. Understanding these sources of error is essential for interpreting the results and assessing the reliability of the system.

Sources of error

Material non-linearity of TPU

TPU does not exhibit perfectly linear elastic behaviour. As the applied force increases, the stiffness of the material changes, leading to a nonlinear force–deformation response. This introduces error in calibration, especially if a simple model is used.

Hysteresis Effect

During loading and unloading, TPU follows different deformation paths. This results in different magnetic field readings for the same force value depending on whether the force is increasing or decreasing, affecting repeatability.

Magnet sensor misalignment

Accurate measurement depends on consistent alignment between the magnet and the sensor. Even small angular or positional misalignments can significantly alter the magnetic field components, leading to inconsistent readings.

External magnetic interference

Nearby electronic devices or metallic objects can disturb the magnetic field, introducing unwanted variations in sensor readings.

Experimental setup errors

Errors may arise from:

- Slight tilting or improper alignment in the UTM setup
- Inconsistent contact between the loading head and sensor
- Delay in stabilization before recording readings

These factors can cause variation in measured data.

Conclusion

In this work, a soft force sensing system based on TPU deformation and magnetic field variation was successfully designed and developed. The sensor utilizes a deformable TPU structure with an embedded triangular magnet, and a magnetometer to detect changes in the magnetic field caused by applied force.

The experimental study demonstrated that applied force leads to measurable deformation of the TPU, which in turn results in a change in the magnetic field detected by the sensor. Due to the nonlinear behaviour of both the material and the magnetic field variation, the relationship between force and sensor output was established through experimental calibration rather than direct analytical modeling.

Through iterative design improvements, issues related to geometry, sensitivity, and alignment were addressed, leading to improved stability and repeatability of the sensor. The final system was able to provide a consistent mapping between magnetic field values and applied force within the tested range.

Although certain limitations such as nonlinearity, hysteresis, and sensitivity to alignment were observed, the results confirm that magnetic field-based soft sensing is a feasible and effective approach for force measurement. The developed system demonstrates the potential of combining soft materials with magnetic sensing techniques for applications requiring flexible and adaptable force sensors.